

About

37 years old
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Benoît Viguiier

PHD. · DOMAIN EXPERT

I am passionate about symmetric cryptography, formal methods, and beautiful code.
I am also a competitive ballroom dancer and a photographer.

Language

French ★★★★★
English ★★★★★
Dutch ★★★★★
German ★★★★★
Spanish ★★★★★
Japanese ★★★★★

OS Preference

Debian ★★★★★
WSL ★★★★★
Windows ★★★★★
MacOs ★★★★★

Activities

Ballroom Dancing
Photography
Sailing
Rock climbing
Piano

Education

PhD in Cryptography & Formal Methods

RADBOD UNIVERSITEIT

Nijmegen, The Netherlands

Sept. 2016 – Dec. 2021

Software Engineer

INSA (NATIONAL INSTITUTE OF APPLIED SCIENCES)

Rennes, France

Sept. 2014 – 2016

MRes. in Computer Science

UNIVERSITY RENNES 1

Rennes, France

Sept. 2015 – 2016

MSc. in Mathematics

UNIVERSITY RENNES 1

Rennes, France

Sept. 2006 – 2011

Skills

Programming

Java, PHP, TypeScript, SQL, Golang, Python, C/C++, Coq, RISC-V asm, ARM Asm

Dev. Env.

VS Code, IntelliJ IDEA, Git, GitHub, AzureDevops, Fortify, NexusLC, SonarQube

Experience

Domain Expert II – Secure Coding

ABN AMRO BANK

Amsterdam, The Netherlands

Oct. 2023 – current

- Member of the Secure Coding team.
- Member of the Quality Assurance board.
- Member of the Cryptographic Working Group.
- Standardization of best practices in software Development.
- Design & implementation of a vulnerability management process with temporal constraints.

Java, Python, TypeScript, Golang, Git, Kubernetes, CICD, Fortify, NexusLC, SonarQube

Domain Expert – Cryptography

ABN AMRO BANK

Amsterdam, The Netherlands

Apr. 2021 – Sep. 2023

- Member of the Crypto Services team.
- Member of the Cryptographic Working Group.
- Design & implementation of the Registration Authority (RA) for the bank Public Key Infrastructure (PKI)
- Certificate Authority migration from Single-tier CA to Two-tier CA.
- Standardization of best practices in cryptography.

Java, PKI, Git, CICD, Agile & Scrum

PhD Researcher

RADBOD UNIVERSITEIT

Nijmegen, The Netherlands

Sep. 2016 – Feb. 2021

- Designing and breaking symmetric cryptography algorithm.
- Writing optimized implementation for lightweight schemes.
- Using formal methods to verify cryptographic C implementations.

Coq, Formal Approaches, Cryptanalysis, C, Assembly

Internship : Software Engineer & Researcher

STMICROELECTRONICS

Brussels, Belgium

Feb. – Jun. 2016

- Verification of the truncated tree search using Formal Methods.
- Application to differential and linear trail search.

Coq, C++, Cryptanalysis, Hash functions

Publications

KangarooTwelve and TurboSHAKE

IRTF — CFRG

INTERNET-DRAFT — RFC-TO-BE 9861

In review

- This document defines the TurboSHAKE128, TurboSHAKE256, KT128 and KT256 eXtendable Output Functions (XOF), i.e., a hash function generalization that can return an output of arbitrary length. Both TurboSHAKE128 and TurboSHAKE256 are based on a Keccak-p permutation specified in [FIPS202] and have a higher speed than the SHA-3 and SHAKE functions.

A Panorama on Classical Cryptography

Nijmegen, The Netherlands

PHD THESIS

Dec. 2021

- In this thesis we cover a large part of the classical cryptography world: we examine the design of new symmetric primitive; we explore implementation strategies of lightweight schemes; we analyze a new high performance algorithm; we use formal verification to prove the correctness of Elliptic Curve Cryptography implementations; and finally we describe one of the way algorithms are standardized.

A Coq proof of the correctness of X25519 in TweetNaCl

Dubrovnik, Croatia

34TH IEEE COMPUTER SECURITY FOUNDATIONS SYMPOSIUM

Jun. 2021

- We formally prove that the C implementation of the X25519 key-exchange protocol in the TweetNaCl library correctly implements the protocol from Bernstein's 2006 paper, as standardized in RFC 7748, as well as the absence of undefined behavior. We also formally prove that X25519 is mathematically correct, i.e., that it correctly computes scalar multiplication on the elliptic curve Curve25519. The proofs are all computer-verified using Coq.

Assembly or Optimized C for Lightweight Cryptography on RISC-V?

Vienna, Austria

CRYPTOLOGY AND NETWORK SECURITY

Dec. 2020

- In this work, we studied the general impact of optimizing symmetric-key algorithms in assembly and in plain C on RISC-V architectures. Additionally, we present optimized implementations of NIST's lightweight candidates, with speed-ups of up to 81% over available implementations, and discuss general implementation strategies.

Cryptanalysis of MORUS

Brisbane, Australia

ADVANCES IN CRYPTOLOGY – ASIACRYPT 2018, LNCS

Dec. 2018

- We present a linear correlation in the keystream of full MORUS, which can be used to distinguish its output from random and to recover some plaintext bits in the broadcast setting.

KangarooTwelve: fast hashing based on Keccak-p

Leuven, Belgium

APPLIED CRYPTOGRAPHY AND NETWORK SECURITY – ACNS 2018, LNCS

July 2018

- KangarooTwelve, a fast and secure arbitrary output-length hash function aiming at a higher speed than the FIPS 202's SHA-3 and SHAKE functions.

Gimli: A Cross-Platform Permutation

Taipei, Taiwan

CRYPTOGRAPHIC HARDWARE AND EMBEDDED SYSTEMS – CHES 2017, LNCS

Sept. 2017

- Gimli, a 384-bit permutation designed to achieve high security with high performance across a broad range of platforms.

Extra Activities

LycheeOrg — Lychee.

The Netherlands

LEAD DEV. & TEAM ADMIN

Aug. 2018 - PRESENT

Lychee is a privacy-friendly self-hosted portfolio gallery designed with a simple look & feel.

- Full rewrite of the PHP core server with Laravel.
- Adopt best industry practices with PHP.
- Support of WebAuthn and OAuth.
- Full rewrite of the front-end from JQuery to Livewire & Alpine in TypeScript.
- Performance analysis and profiling.
- Full rewrite of the front-end to Vuejs 3 in TypeScript.

Landscapes & Ballroom Photography

The Netherlands

PHOTOGRAPHER

Jul. 2018 - PRESENT

- Landscapes & Astrophotography
- Ballroom Photography – WDSF, NADB, NTDS, ETDS
- Events – PhD Defenses
- Portraiture

Standard Ballroom & Formation Dancer

The Netherlands

Mar. 2018 - Dec. 2024

MEMBER OF DSV SWAY OF LIFE

- 2024 — **3rd place** — World Championship in Formation Dancing — Vienna
- 2023 — **5th place** — Dutch Championship in couple S-class – Senior 1 — Steenwijk
- 2023 — **1st place** — Dutch Championship in Formation Dancing — Rotterdam
- 2023 — **3rd place** — Dutch Championship in couple A-class — Steenwijk
- 2022 — **1st place** — Dutch Championship in Formation Dancing — Dalfsen
- 2022 — **5th place** — World Championship in Formation Dancing — Braunschweig
- 2022 — **3rd place** — European Championship in Formation Dancing — Nürnberg
- 2022 — **Semi-Finalist** — Dutch Championship in couple A-class — Steenwijk
- 2021 — **2nd place** — Dutch Championship in couple C-class — Dalfsen
- 2021 — **1st place** — Dutch Championship in Formation Dancing — Rotterdam
- 2020 — **4th place** — Dutch Championship in couple C-class — Dalfsen
- 2019 — **1st place** — Dutch Championship in Formation Dancing — Almere
- 2019 — **Finalist** — World Championship in Formation Dancing — Moskow
- 2018 — **1st place** — Dutch Championship in Formation Dancing — Almere